

GigScore: Master Presentation Deck & Architecture Guide

Comprehensive AI Prompt, Slide-by-Slide Blueprint, System Design, and Technical Specifications

Project: GigScore Alternative Credit Platform	Target Domain: FinTech / Gig-Economy Lending
Architecture: FastAPI + MongoDB + React 19 + OpenCV	ML Core: Calibrated XGBoost v3.2 + SHAP TreeExplainer
Concurrency: 50+ Concurrent Active Users (Zero Drops)	Document Purpose: Prompt for PPT Generators & Deck Script

1. Master Copy-Paste Prompt for AI Presentation Tools

Copy and paste the exact text below into AI presentation tools such as **Gamma App**, **ChatGPT (with SlidesGPT)**, **Claude**, **Tome**, or **Beautiful.ai** to automatically generate the full presentation.

Act as an elite FinTech Systems Architect, ML Engineer, and Technical Presenter. Generate a comprehensive, 14-slide, professional presentation deck for our project: 'GigScore: Explainable Alternative Credit Scoring & Underwriting Platform for Gig-Economy Workers'.

Create modern, high-impact, content-rich slides with clear visual hierarchies, system architecture diagrams (ASCII/Mermaid), quantitative ML metrics, step-by-step operational workflows, and speaker notes for every slide.

Ensure the presentation thoroughly covers:

1. **The Problem:** The credit invisibility crisis of 15M+ gig drivers & delivery partners (lack of CIBIL/bureau scores, volatile weekly cash flows, predatory loan shark traps).
2. **The Solution:** Alternative underwriting using earnings telemetry, ride data, behavioral consistency, biometric KYC, and explainable AI.
3. **System Architecture & Concurrency:** FastAPI ASGI cluster, AnyIO threadpool scaling (150 worker tokens), MongoDB Atlas connection pool (min=15, max=150), Cloudinary CDN, and pre-warmed models.
4. **Step-by-Step Workflow:** Onboarding -> Statement ingestion -> Parsing -> 18 Feature Engineering -> Calibrated ML scoring (300-900) -> SHAP waterfall -> Underwriter sanction cockpit -> Disbursement.
5. **Machine Learning & Explainability:** Calibrated XGBoost v3.2 (ROC-AUC 0.957, PR-AUC 0.922, Accuracy 93.25%, Brier score 0.061), feature importance ranking, and SHAP TreeExplainer local attribution.
6. **Biometric Face Security:** OpenCV DNN YuNet (5-point microsecond landmarks) + SFace (128-D deep embeddings) + geometric facial invariant analysis.
7. **Tri-Portal UX:** Driver Portal, Lender / Underwriter Cockpit, and Admin Ops / MLOps Console.
8. **Production Engineering:** Sub-second P95 latency (<450ms), GZip compression, and anti-sleep keepalive background bot.

Follow the detailed 14-slide outline below:

- Slide 1: Title Slide (Cover, Metadata, Vision)
- Slide 2: The Core Problem & Market Opportunity (Credit Invisibility Trap)
- Slide 3: The GigScore Solution & Core Value Proposition
- Slide 4: High-Level System Architecture & Concurrency Design
- Slide 5: Step-by-Step Operational Lifecycle (End-to-End Workflow)
- Slide 6: Feature Engineering Pipeline (The 18 Core Signals)
- Slide 7: Machine Learning Benchmark & Model Competition
- Slide 8: Explainable AI (XAI) & SHAP Transparency Engine
- Slide 9: Biometric Face Authentication & Anti-Fraud Computer Vision
- Slide 10: Multi-Role System Portals & User Experience (3 Cockpits)
- Slide 11: Underwriting Rules & Automated Policy Decision Matrix
- Slide 12: Production Engineering, Resilience & Latency Benchmarks
- Slide 13: Horizontal & Vertical Scalability Roadmap
- Slide 14: Conclusion, Business Impact & Strategic Vision

2. System Design & Concurrency Architecture Specification

GigScore is engineered with a high-concurrency architecture capable of supporting **50+ simultaneous active users** with zero request drops, sub-second latency (P95 < 450ms), and full transactional integrity.



Component	Default Setting	GigScore High-Capacity Tuned	Impact Solved
AnyIO Threadpool	40 worker tokens	150 worker tokens	Eliminates thread exhaustion under 50-user burst traffic.
MongoDB Sockets	minPoolSize = 0	minPoolSize = 15, max = 150	Zero TLS renegotiation latency spikes.
Model Lifecycle	Load on first request	Pre-warmed in FastAPI lifespan	Prevents cold-start disk I/O thundering herd.
Payload Transfer	Uncompressed JSON	GZipMiddleware (min: 1KB)	Reduces SHAP factor download sizes by 75%.

3. Step-by-Step Project Workflow (End-to-End)

Here is how a transaction travels from the driver's smartphone through the data processing pipeline to the lender's loan sanctioning desk:

Step 1: Driver Onboarding & Biometric KYC Enrollment

The driver signs up via the Driver Portal. The system initiates a browser webcam session where the driver's face is scanned. The OpenCV YuNet deep detector pinpoints 5 key landmarks, and SFace computes an affine-aligned 128-dimensional biometric vector. The portrait is stored securely on Cloudinary CDN, and the biometric embedding is persisted to the MongoDB user document.

Step 2: Platform Earnings Statement Ingestion

The driver uploads their earnings statements (PDF/CSV/DOCX) from platforms like Uber, Ola, Swiggy, or Zomato. The file is validated, uploaded to Cloudinary CDN, and recorded in MongoDB with rollback protection.

Step 3: Multi-Format Document Parsing & Normalization

The PdfParserService and DocumentParserService run regex and tabular extraction pipelines. It extracts monthly trips, active days, gross earnings, platform commission deductions, fuel costs, net income, cancellation rates, and customer ratings.

Step 4: 18-Dimensional Feature Engineering

The FeatureService transforms raw monthly telemetry into 18 ML variables: lowest monthly income floor (min_income), income volatility (coefficient of variation), 3-month income trajectory slope, disposable income surplus, and requested EMI-to-income ratio.

Step 5: Calibrated XGBoost Inference & Credit Scoring

The pre-warmed calibrated XGBoost model predicts the exact Probability of Default (Pd). The system transforms Pd into an institutional credit score: **Score = 900 - (Pd * 600)**, bounded between 300 and 900. The score is classified into Risk Tiers: Prime (≥ 750), Near-Prime (600-749), or Subprime (< 600).

Step 6: SHAP Explainability & Factor Attribution

The SHAP TreeExplainer calculates local feature attributions around the base expected score. The system generates human-readable explanations (e.g., 'Low cancellation rate boosted score by +32 points') and actionable tips.

Step 7: Underwriter Sanction Cockpit & Policy Decisioning

The loan application enters the Underwriter review queue with status PENDING. The underwriter examines the score, risk band, debt-serviceability ratio, and SHAP factors, and can Approve, Reject, or Counter-Offer with adjusted terms.

Step 8: Sanction Generation & Disbursement

Upon sanction approval, an automated sanction agreement is logged in the MongoDB audit trail, the driver is notified on their dashboard, and the repayment schedule is initialized.

4. Complete 14-Slide Presentation Deck Blueprint

This section provides the exact content, visual design recommendations, and speaker scripts for every single slide in the deck.

SLIDE 1: Title Slide (Cover)

Subtitle: Unlocking Fair, Data-Driven Lending for the Gig Economy

Visual / Layout: Dark navy theme (#0B132B), emerald accent badge, dual driver silhouette and interactive credit score gauge.

Key Slide Content:

- **Project:** GigScore Alternative Credit Risk Assessment Platform
- **Core Capabilities:** Telemetry Underwriting | Calibrated Machine Learning | SHAP XAI | Face Biometrics
- **Target Sector:** 15 Million+ underserved gig workers, taxi drivers, and delivery partners across emerging markets
- **Architectural Scale:** High-concurrency engine engineered for 50+ simultaneous active users with zero request drops

***Presenter Talking Point:** "Introduce the team and hook the audience: Gig workers power our modern cities, yet when they apply for formal loans, banks shut the door. GigScore changes this."*

SLIDE 2: The Core Problem & Market Opportunity

Subtitle: The Credit Invisibility Trap for 15M+ Gig Workers

Visual / Layout: Three-way comparison chart: Traditional Banking Requirements vs. Gig Worker Reality vs. Harmful Outcomes.

Key Slide Content:

- **Bureau Blindspot:** Over 87% of gig workers have no CIBIL or bureau history; traditional algorithms auto-reject them due to lack of payslips and Form 16.
- **Income Volatility Misinterpretation:** Drivers earn steady annualized income (Rs. 25,000 to Rs. 50,000/mo), but weekly volatility triggers false credit risk flags.
- **Predatory Debt Traps:** Workers are forced into loan shark agreements charging 60% to 120% APR for routine vehicle maintenance or medical emergencies.
- **The Multi-Billion Market:** Massive untapped lending opportunity for FinTech NBFCs if underwriting can be grounded in actual ride telemetry.

***Presenter Talking Point:** "Stress that gig workers are not credit-unworthy—they are simply credit-invisible under outdated banking frameworks."*

SLIDE 3: The GigScore Solution & Core Value Proposition

Subtitle: Transforming Driving Telemetry into Verifiable Creditworthiness

Visual / Layout: Central engine graphic taking in 4 platform streams and outputting an institutional 300–900 GigScore.

Key Slide Content:

- **Holistic Alternative Underwriting:** Converts platform statements (trips, earnings, fuel, cancellation rates) into bankable credit profiles.

- **Institutional 300 to 900 Score:** Calibrated probability-of-default mapped directly to familiar credit tiers (Prime, Near-Prime, Subprime).
- **Explainable AI (XAI) by Design:** Zero black-box rejections; SHAP TreeExplainer delivers exact reasons and actionable improvement advice.
- **Deep Biometric KYC:** Microsecond OpenCV YuNet + SFace facial verification prevents identity theft and unauthorized account sharing.
- **Sub-Second Sanctioning:** P95 latency under 450ms delivers instantaneous pre-qualification and automated underwriter routing.

***Presenter Talking Point:** "Walk through the four pillars: Telemetry data, Calibrated ML, Explainable AI, and Deep Biometric Security."*

SLIDE 4: System Design: High-Concurrency Cloud Architecture

Subtitle: Engineered for 50+ Concurrent Users & Sub-Second Latency

Visual / Layout: Tiered architecture diagram showing Client -> Ingress -> AnyIO Threadpool -> Pre-Warmed ML/Vision Engines -> MongoDB Atlas.

Key Slide Content:

- **Client Tier:** React 19 + Vite responsive SPA with dedicated driver, underwriter, and MLOps dashboards.
- **Ingress Tier:** Uvicorn ASGI cluster with 2048 backlog and GZip payload compression reducing network traffic by up to 75%.
- **Concurrency Scaler:** Starlette AnyIO threadpool expanded from 40 to 150 worker tokens (600 total across workers), supporting 100 RPS bursts.
- **Pre-Warmed Engine Lifespans:** XGBoost, SHAP explainer, and OpenCV models pre-warmed into RAM—zero cold-start file I/O delays.
- **Database Connection Pool:** MongoDB Atlas configured with minPoolSize=15 pre-connected TCP sockets, eliminating TLS handshake spikes.

***Presenter Talking Point:** "Emphasize to technical judges that this is not a prototype script—the concurrency and memory math were explicitly engineered for production workloads."*

SLIDE 5: Step-by-Step Operational Lifecycle

Subtitle: The Complete Journey from Document Upload to Loan Sanction

Visual / Layout: 8-phase horizontal chevron pipeline showing real-time data transformations.

Key Slide Content:

- **1. Driver Biometric Enrollment:** Webcam capture, YuNet landmark detection, and SFace 128-D vector saved to MongoDB & Cloudinary.
- **2. Statement Upload:** PDF/CSV earnings statements uploaded with automatic format validation and rollback protection.
- **3. Document Parsing:** Regex and tabular parsers extract monthly trips, gross earnings, platform cuts, fuel costs, and cancellation rates.
- **4. Feature Synthesis:** Generates 18 telemetry features including disposable income, coefficient of variation, and income trends.
- **5. Calibrated Scoring:** Calibrated XGBoost predicts default probability and computes the 300–900 GigScore.
- **6. SHAP Transparency:** Computes local feature contributions for driver-facing tips and underwriter risk audits.
- **7. Underwriter Review:** Underwriter inspects application in Lender Cockpit and selects Approve, Counter-Offer, or Deny.

- **8. Sanction & Disbursement:** Sanction letter generated and recorded in immutable MongoDB audit logs.

Presenter Talking Point: "Explain that every step is synchronous, auditable, and completed in under 2 seconds total."

SLIDE 6: Feature Engineering: The 18 Multidimensional Signals

Subtitle: Extracting Predictive Behavioral Signals from Unstructured Telemetry

Visual / Layout: 4-quadrant diagram grouping features into Income Stability, Operational Discipline, Platform Tenure, and Affordability.

Key Slide Content:

- **Income Volatility Group:** min_income (earnings safety floor), avg_monthly_net_income, income_std, coefficient_of_variation (CV = sigma / mu), months_with_income.
- **Operational Discipline Group:** cancellation_rate (key proxy for reliability), completion_rate, avg_rating (customer service index).
- **Platform Tenure & Intensity:** tenure_months, active_days_monthly, trips_per_day, trips_per_month.
- **Dynamic Earning Patterns:** peak_hour_share, weekend_share, income_slope_3m (3-month rolling growth rate), recent_vs_historical_income.
- **Affordability Guardrails:** estimated_disposable_income, requested_emi_to_income.

Presenter Talking Point: "Point out how features like cancellation_rate and min_income capture operational conscientiousness and worst-case cash flow resilience."

SLIDE 7: Machine Learning Benchmarks & Model Competition

Subtitle: Champion XGBoost Model Outperforms Baselines with 93.25% Accuracy

Visual / Layout: Model benchmark table with highlighted champion row and horizontal bar chart of top 5 feature importances.

Key Slide Content:

- **Model Evaluation Suite:** Evaluated Logistic Regression (Baseline), Random Forest, and Calibrated XGBoost across 2,000 samples.
- **Champion Performance:** Calibrated XGBoost achieved **ROC-AUC: 0.957**, **PR-AUC: 0.922**, **Accuracy: 93.25%**, and **Brier Score: 0.061**.
- **The Calibration Advantage:** Platt sigmoid calibration aligns raw tree margins with true empirical default probabilities.
- **Top Feature Importances:**
 - min_income: **48.0%** (Driver's baseline safety floor)
 - cancellation_rate: **11.6%** (Operational reliability)
 - coefficient_of_variation: **10.2%** (Income stability factor)
 - completion_rate: **7.0%** (Trip fulfillment commitment)

Presenter Talking Point: "Highlight that min_income and cancellation_rate drive nearly 60% of the model's predictive power—demonstrating that behavioral discipline correlates with creditworthiness."

SLIDE 8: Explainable AI (XAI) & SHAP Transparency Engine

Subtitle: Transforming Opaque Risk Scores into Actionable Driver Coaching

Visual / Layout: Mockup of SHAP waterfall attribution plot and driver-facing credit improvement card.

Key Slide Content:

- **The Black-Box Dilemma:** Fair lending regulations and RBI mandates reject opaque credit decisions without actionable adverse action notices.
- **SHAP TreeExplainer Formulation:** Decomposes each individual score into additive contributions: $GigScore(x) = E[f(x)] + \sum(\phi_i)$.
- **Driver Empowerment:** Drivers see exact reasons: 'High customer rating (+28 pts)', 'Zero cancellations in August (+35 pts)', 'Income volatility (-42 pts)'.
- **Actionable Coaching:** Provides concrete advice: 'Reducing ride cancellations below 4% will increase your GigScore by +38 points, unlocking lower interest rates.'

***Presenter Talking Point:** "Explain how explainability builds trust. Instead of feeling cheated by an algorithm, drivers are coached on how to improve."*

SLIDE 9: Biometric Face Authentication & Anti-Fraud Vision

Subtitle: Microsecond OpenCV DNN Face Recognition & Invariant Geometry

Visual / Layout: Facial mesh diagram showing YuNet 5-point landmarks, SFace affine alignment, and 128-D vector cosine similarity matching.

Key Slide Content:

- **OpenCV YuNet Detection:** Microsecond deep face detection and 5-point landmark localization (eyes, nose, oral commissures) at 320x320 resolution.
- **OpenCV SFace 128-D Vectors:** Landmark-guided affine alignment normalizes head tilt before computing deep 128-dimensional metric embeddings.
- **Geometric Invariant Analysis:** Multi-element geometry verifies inter-ocular distance, golden triangles, and nasal-oral symmetry ratios.
- **Anti-Fraud & Privacy:** Calibrated threshold (0.55) separates genuine users (>0.88) from impostors (<0.42). Video is never stored; only encrypted vectors saved in MongoDB.

***Presenter Talking Point:** "Explain that biometric security prevents common gig-economy fraud like account renting and loan stacking under fake identities."*

SLIDE 10: Multi-Role User Experience: Three Purpose-Built Cockpits

Subtitle: Streamlined Interfaces for Drivers, Underwriters, and Platform Admins

Visual / Layout: Three mockups showing Driver Score Gauge, Underwriter Review Cockpit, and Admin System Monitoring Console.

Key Slide Content:

- **1. Driver Portal:** Interactive 300–900 Credit Score Gauge, risk tier badges, earnings volatility charts, and 1-click loan requests.
- **2. Lender & Underwriter Cockpit:** Live pending application queue, underwriting sanction speedometer, 1-click approval/counter-offer, and SHAP factor deep dive.
- **3. Admin & MLOps Console:** Real-time AnyIO thread capacity, MongoDB connection pool health, model drift tracking, and immutable audit logs.

Presenter Talking Point: "Mention that role-based access control (RBAC) ensures strict data isolation between drivers and institutional underwriters."

SLIDE 11: Underwriting Policies & Risk Band Decision Matrix

Subtitle: Automated Guardrails Balancing Rapid Sanctions with Capital Protection

Visual / Layout: Three-tiered risk table with color-coded policy rules and affordability formulas.

Key Slide Content:

- **Credit Score Formula:** $\text{GigScore} = 900 - (\text{Probability_of_Default} * 600)$, clamped between 300 and 900.
- **Prime Tier (Score ≥ 750 / Low Risk):** Auto-eligible up to Rs. 3,00,000 (up to 6x disposable income). EMI affordability ratio capped at 35%.
- **Near-Prime Tier (Score 600-749 / Medium Risk):** Eligible up to Rs. 1,20,000 (up to 3.5x disposable income). Routed to Underwriter Manual Review.
- **Subprime Tier (Score < 600 / High Risk):** Loan capped at Rs. 35,000 (1x disposable income). Requires risk mitigation.
- **Affordability Guardrail:** $\text{Affordability Ratio} = \text{Requested EMI} / \text{Estimated Disposable Income}$. Prevents over-leveraging regardless of score.

Presenter Talking Point: "Show how the policy engine protects lenders by capping loans strictly to disposable income surplus, not just top-line gross earnings."

SLIDE 12: Production Engineering, Resilience & Latency Benchmarks

Subtitle: Sub-Second P95 Latency and Fault-Tolerant Cloud Infrastructure

Visual / Layout: Metrics grid displaying P95 latency ($< 450\text{ms}$), 0% drop rate, 75% compression ratio, and 50+ concurrent capacity.

Key Slide Content:

- **Sub-Second P95 Latency:** Complete credit evaluation and SHAP factor extraction executed in under 450ms.
- **High-Concurrency Scalability:** AnyIO threadpool scaled to 150 worker tokens per process easily absorbs 100 RPS traffic spikes.
- **Bandwidth Optimization:** GZip compression reduces large JSON explainability trees and audit records by up to 75%.
- **Keepalive Background Daemon:** Integrated self-ping service and UptimeRobot hooks pinging every 5 minutes prevent cloud hosting sleep cycles.
- **Transactional Rollback Protection:** Document uploads clean up orphaned Cloudinary assets if MongoDB persistence fails.

Presenter Talking Point: "Highlight that system reliability is baked in at every tier—from the network layer to the database connection pool."

SLIDE 13: Scalability Roadmap & Enterprise Expansion

Subtitle: From 50 Concurrent Users to National Platform Deployment

Visual / Layout: 3-phase evolution roadmap: Phase 1 (Single Node) -> Phase 2 (Distributed Cluster) -> Phase 3 (Kubernetes & Open Banking).

Key Slide Content:

- **Phase 1 (Current Target):** 50 Concurrent Active Users | 1 Node, 4 Uvicorn workers, 150 AnyIO threads, Mongo Atlas M0/M2.
- **Phase 2 (Growth Target):** 250 Concurrent Users | 2 Nodes behind AWS Application Load Balancer, Redis session cache, Mongo Atlas M10.
- **Phase 3 (National Scale):** 1,000+ Concurrent Users | Kubernetes Horizontal Pod Autoscaler (HPA), Celery/RabbitMQ distributed OCR workers.
- **Open Banking Integration:** Direct API integration with India's Account Aggregator (AA) framework (Sahamati/Setu) for instant, consent-based bank statement pulls.

Presenter Talking Point: "Show the vision forward: GigScore is modular and ready to plug directly into the national Account Aggregator ecosystem."

SLIDE 14: Conclusion: Fair Credit for the Gig Economy

Subtitle: Financial Inclusion Powered by Intelligent Engineering

Visual / Layout: Inspiring closing layout with summary metrics, partnership vision, and Q&A; prompt.

Key Slide Content:

- **Empowering the Hardworking:** Converts daily hustle on the road into verifiable, institutional-grade creditworthiness.
- **De-Risking Digital Lenders:** 93.25% ML accuracy, calibrated risk bands, and biometric verification minimize non-performing loans.
- **Built for Production:** Complete, working codebase with FastAPI backend, MongoDB Atlas, React 19 frontend, and OpenCV vision.
- **Strategic Vision:** 'GigScore: Where Gig Work Meets Fair, Transparent Credit.'
- **Q&A; Session:** Thank you! We are now ready for questions and a live platform demonstration.

Presenter Talking Point: "Deliver a strong, memorable closing and invite the evaluators or audience to ask questions or watch the live demo."